

SentinelWire™ v1.1 — Instruction Manual

(How to Use the Framework End-to-End)

Purpose

SentinelWire™ is a **Wireshark-driven SOC investigation framework** designed to help analysts move from **raw packet captures** to **structured, explainable security findings**.

Version **v1.1** extends SentinelWire with **behavior-based detection**, enabling analysts to identify **command-and-control beaconing** and **lateral movement** using short PCAP captures.

Step-by-Step Workflow

Step 1 — Install & Personalize Profiles

Run the profile installer:

```
make_ws_profile.bat
```

Provide:

- Your workstation IP
- Router / default gateway IP
- Your MAC address
- Router MAC address

This installs and personalizes all SentinelWire Wireshark profiles under:

```
%APPDATA%\Wireshark\profiles\
```

Step 2 — First-Click Capture (Triage)

1. Open Wireshark
2. Select **UNIVERSAL-DEFENSE** profile
3. Start capture on the active interface
4. Capture **30–60 seconds only**
5. Stop capture

This capture is intentionally short to:

- Reduce noise

- Preserve analyst focus
- Emphasize behavioral signals

Step 3 — Run IOC Extraction

Execute the updated extractor:

```
powershell -ExecutionPolicy Bypass -File extract_iocs_UPDATED_v1.1.ps1 -PcapFile capture.pcapng
```

A new run folder is created automatically containing:

- ioc_scored.csv
- ioc_scored_top20.txt
- ioc_scored_summary.txt

Step 4 — Review Results

Start with:

ioc_scored_summary.txt

Then review:

ioc_scored_top20.txt

These files are designed for **fast analyst decision-making**, not raw data exploration.

Step 5 — Pivot to Specialized Profiles

Based on detected signals:

- Beaconsing → **BEACON-TIME**
- Lateral movement → **SMB-LATERAL, LLMNR-SUSPECT**
- DNS anomalies → **DNS-SUSPECT, LONG-DNS**
- MITM signals → **MITM-MASTER, ARP-SPOOF**